Calorimetry & Latent Heat

Student Notes + Worked Problems + Practice

Main formulas

$$Q_{\text{lost}} = Q_{\text{gained}}, \qquad Q = mc\Delta T, \qquad Q = mL$$

Use $Q = mc\Delta T$ when temperature changes without a phase change. Use $Q = mL$ when the phase changes at constant temperature.

1. Principle of Calorimetry

1.1 Heat lost = Heat gained

When a hot body is placed in contact with a colder body, heat flows from the hotter body to the colder body until both reach the same final temperature.

For an insulated system,

$$\boxed{Q_{\text{lost}} = Q_{\text{gained}}}$$

This is simply conservation of energy.

1.2 Writing heat lost and heat gained

For a hot body cooling from T_h to T_f:

$$Q_{\text{lost}} = m_h c_h (T_h - T_f)$$

For a cold body warming from T_c to T_f:

$$Q_{\text{gained}} = m_c c_c (T_f - T_c)$$

Hence,

$$\boxed{m_h c_h (T_h - T_f) = m_c c_c (T_f - T_c)}$$

Safe temperature rule

$$\Delta T = \text{higher temperature} - \text{lower temperature}$$

This keeps both heat lost and heat gained positive.

1.3 Final equilibrium temperature

If no phase change occurs, the final temperature must lie between the two starting temperatures:

$$\boxed{T_c < T_f < T_h}$$

If the answer lies outside this range, check the calculation.

Worked Example 1 - Mixing water

10 g of water at 70°C is mixed with 5 g of water at 30°C. Find the final temperature.
Since both are water, $c = 1\,\text{cal/(g}\,^\circ\text{C)}$:

$$10(1)(70 - T) = 5(1)(T - 30)$$

$$700 - 10T = 5T - 150$$

$$850 = 15T$$

$$\boxed{T = 56.7^\circ\text{C}}$$

The answer is closer to 70°C because there is more hot water.

1.4 Mixing different materials

The same principle works even when the two bodies have different specific heats.

Worked Example 2 - Metal + water

A 100 g metal block of specific heat $0.2\,\text{cal/(g}\,^\circ\text{C)}$ at 100°C is placed in 200 g of water at 20°C. Find the final temperature.

$$100(0.2)(100 - T) = 200(1)(T - 20)$$

$$20(100 - T) = 200(T - 20)$$

$$2000 - 20T = 200T - 4000$$

$$6000 = 220T$$

$$\boxed{T \approx 27.3^\circ\text{C}}$$

1.5 Meaning of mc

From

$$Q = mc\Delta T$$

we get

$$\boxed{\Delta T = \frac{Q}{mc}}$$

Therefore,

$$\boxed{\text{large } mc \Rightarrow \text{small temperature change}}$$

$$\boxed{\text{small } mc \Rightarrow \text{large temperature change}}$$

So, in a mixing problem, the final temperature tends to stay closer to the initial temperature of the body with the larger mc.

2. Latent Heat

2.1 Why temperature stays constant during a phase change

Suppose ice at 0°C is heated. The ice starts melting, but the temperature stays at 0°C until all the ice has melted.

The supplied heat can do two different jobs:

1. Increase molecular motion ⇒ kinetic energy increases ⇒ temperature rises.
2. Change molecular arrangement ⇒ potential energy increases ⇒ phase changes while temperature stays constant.

During a phase change, heat is used to change the state of the substance rather than raise its temperature.

2.2 Latent heat and $Q = mL$

Latent heat L is the heat required to change the state of unit mass of a substance at constant temperature.

$$L = \frac{Q}{m}$$

Therefore,

$$\boxed{Q = mL}$$

Situation	Formula
Temperature changes, phase remains the same	$Q = mc\Delta T$
Phase changes, temperature remains constant	$Q = mL$

2.3 Fusion and freezing

For water,

$$\boxed{L_f = 80\,\text{cal/g}}$$

Melting:

$$\text{ice at } 0^\circ\text{C} \rightarrow \text{water at } 0^\circ\text{C}$$

Heat absorbed:

$$Q = mL_f$$

Freezing is the reverse process and releases the same amount of heat.

2.4 Vaporisation and condensation

For water,

$$\boxed{L_v \approx 540\,\text{cal/g}}$$

Boiling:

$$\text{water at } 100^\circ\text{C} \rightarrow \text{steam at } 100^\circ\text{C}$$

Heat absorbed:

$$Q = mL_v$$

Condensation is the reverse process and releases the same amount of heat.

2.5 Why vaporisation needs much more energy

$$\frac{L_v}{L_f} = \frac{540}{80} \approx 6.7$$

Thus,

$$\boxed{L_v \approx 6.7 L_f}$$

Boiling requires much more energy because the molecules must separate much more strongly than during melting.

2.6 The five-rung ice-to-steam ladder

For a complete heating process, always split the journey into stages.

Stage	Process	Heat
1	Warm ice to 0°C	$Q_1 = mc_{\text{ice}}\Delta T$
2	Melt ice at 0°C	$Q_2 = mL_f$
3	Warm water to 100°C	$Q_3 = mc_{\text{water}}\Delta T$
4	Boil water at 100°C	$Q_4 = mL_v$
5	Warm steam above 100°C	$Q_5 = mc_{\text{steam}}\Delta T$

$$\boxed{Q_{\text{total}} = Q_1 + Q_2 + Q_3 + Q_4 + Q_5}$$

Best method: draw all five stages first, then cross out the stages that are not required in the question.

Worked Example 3 - Two-stage ladder

How much heat is needed to convert 20 g of ice at 0°C into water at 40°C?
Melting:

$$Q_1 = 20(80) = 1600\,\text{cal}$$

Heating water:

$$Q_2 = 20(1)(40) = 800\,\text{cal}$$

Total:

$$\boxed{Q = 1600 + 800 = 2400\,\text{cal}}$$

Worked Example 4 - Full ladder to steam

How much heat is needed to convert 10 g of ice at -20°C completely into steam at 100°C?
Take

$$c_{\text{ice}} = 0.5, \qquad c_{\text{water}} = 1,$$
$$L_f = 80\,\text{cal/g}, \qquad L_v = 540\,\text{cal/g}.$$

Warm the ice:

$$Q_1 = 10(0.5)(20) = 100\,\text{cal}$$

Melt the ice:

$$Q_2 = 10(80) = 800\,\text{cal}$$

Warm the water:

$$Q_3 = 10(1)(100) = 1000\,\text{cal}$$

Boil the water:

$$Q_4 = 10(540) = 5400\,\text{cal}$$

Therefore,

$$\boxed{Q = 100 + 800 + 1000 + 5400 = 7300\,\text{cal}}$$

2.7 Finding the final state when heat supplied is given

Sometimes the question gives a fixed amount of heat but does not tell you the final state. In such questions, use the available heat stage by stage.

Worked Example 5 - Final-state problem

10 g of ice at 0°C is supplied 2000 cal. Find the final state and temperature.
First melt all the ice:

$$Q = 10(80) = 800\,\text{cal}$$

Heat remaining:

$$2000 - 800 = 1200\,\text{cal}$$

Now heat 10 g of water from 0°C to 100°C:

$$Q = 10(1)(100) = 1000\,\text{cal}$$

Heat remaining:

$$1200 - 1000 = 200\,\text{cal}$$

The remaining 200 cal converts some water to steam:

$$200 = m(540)$$

$$m = \frac{200}{540} \approx 0.37\,\text{g}$$

Therefore,

$$\boxed{\text{steam} = 0.37\,\text{g}}$$

$$\boxed{\text{water} = 9.63\,\text{g}}$$

$$\boxed{T = 100^\circ\text{C}}$$

If only part of a substance changes phase, the temperature remains at the phase-change temperature while both phases are present.

3. Quick Revision

$$\boxed{Q_{\text{lost}} = Q_{\text{gained}}}$$

$$\boxed{Q = mc\Delta T \quad \text{for temperature change}}$$

$$\boxed{Q = mL \quad \text{for phase change}}$$

$$\boxed{L_f = 80\,\text{cal/g}, \qquad L_v \approx 540\,\text{cal/g}}$$

Warm ice → Melt → Warm water → Boil → Warm steam

4. Practice Problems

Problem 1 - Calorimetry

Equal masses of two liquids are initially at 20°C and 80°C. After mixing, the final temperature is 40°C. Find

$$\frac{c_{\text{cold}}}{c_{\text{hot}}}.$$

Problem 2 - Three-stage ladder

Find the heat required to convert 25 g of ice at -10°C into water at 30°C.

Take

$$c_{\text{ice}} = 0.5\,\text{cal/(g}\,^\circ\text{C)}, \qquad c_{\text{water}} = 1\,\text{cal/(g}\,^\circ\text{C)},$$
$$L_f = 80\,\text{cal/g}.$$

Problem 3 - Reverse ladder

5 g of steam at 100°C is converted completely into ice at 0°C. Find the heat released.

Take

$$L_v = 540\,\text{cal/g}, \quad c_{\text{water}} = 1\,\text{cal/(g}\,^\circ\text{C)}, \quad L_f = 80\,\text{cal/g}.$$

Problem 4 - Final state

5 g of ice at 0°C is supplied 600 cal. Find the final state and temperature.

Problem 5 - Challenge

20 g of ice at -10°C is supplied 2500 cal. Find the final state and temperature.

Take

$$c_{\text{ice}} = 0.5\,\text{cal/(g}\,^\circ\text{C)}, \qquad c_{\text{water}} = 1\,\text{cal/(g}\,^\circ\text{C)}, \qquad L_f = 80\,\text{cal/g}.$$

5. Answers to Practice Problems

1. $\boxed{\dfrac{c_{\text{cold}}}{c_{\text{hot}}} = 2}$
2. $\boxed{2875\,\text{cal}}$
3. $\boxed{3600\,\text{cal released}}$
4. $\boxed{5\text{ g water at }40^\circ\text{C}}$
5. Heat to warm ice to 0°C: $20(0.5)(10) = 100$ cal. Heat to melt all ice: $20(80) = 1600$ cal. Remaining heat: $2500 - 1700 = 800$ cal. This heats 20 g water by

$$800 = 20(1)\Delta T \Rightarrow \Delta T = 40^\circ\text{C}.$$

So the final state is $\boxed{20\text{ g water at }40^\circ\text{C}}$.